

Improvement of Warning Lag Time in Forward Collision Warning Systems based on Multifunctional Warnings

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Abstract— We present an improvement of forward collision warning (FCW) by adding more layers of warning functionality based on five stages of crash timeline. Generally, FCW operates at stage 3, the critical moment before crash. The added functionality is the location-based warning (LW) which warns the driver when driving toward a dangerous spot with high speed. LW operates at stage 2, an open condition to crash which comes from bad driving habit. We propose the use of warning lag time as a performance indicator, which is appropriate for real driving experiment. The warning lag time is defined as the time period between the issue of warning and the execution of brake by a normal driver. We performed real driving experiments with normal drivers and compare their brake actions with warnings from our system. The system with FCW alone achieved an average time lag of 1.47 sec., while the system with both FCW+LW can warn the driver faster with average time lag of 1.32 sec. The accuracy of both systems was about the same at 79%.

I. INTRODUCTION

In many traffic accidents, the damage to lives and properties is so severe that it cannot be estimated. This problem brings about the development of safety systems in order to reduce both accidents and injuries. One common cause of vehicle accidents is driver's inattentiveness to the driving task because of interrupting calls or readiness of his driving conditions such as fatigue or sleepiness. Since the major cause of accidents comes from human errors, active safety and intelligent driving assistance system are necessary solutions for future transportation.

The forward collision warning (FCW) is an emerging technology that has been developed to help drivers avoid collisions by giving warnings in dangerous situations. FCW systems usually implement variety of sensors in order to measure the distance to the front cars, the velocity, and the acceleration of the equipped cars [1], [2]. Then, these measured quantities are used in a mathematic model of vehicles' motion to predict potential crashes. In general, forward collision warning signals can be issued within the last few seconds before crash, called the critical moment. This may not be enough for sluggish drivers to avoid a crash in tight situations. Therefore, there are many researchers who made efforts to predict car crashes and increase time period between warning and crash called pre-warning time. Causes

of frontal collision, such as following distance, speed, acceleration, deceleration, road environment, etc., are considered as parameters to enhance FCW system. Masumi et al. [3] included braking reaction time of a driver and a relative safety distance margin in the vehicle's motion model in order to increase the pre-warning time. The vehicle deceleration is also a significant parameter of collision model. Deceleration for collision avoidance (DCA) is calculated and used as an indicator of forward collision risk [4], [5]. Another interesting method to enhance FCW systems is a combination method. Road environment detection, such as lane departure, and frontal obstacles, is utilized to improve the performance of FCW systems. Chih-Li Hou et al. [6] integrated an FCW system together with a lane departure warning system based on image processing and the vehicle kinematic model. The mentioned work accomplished improvement of FCW systems in term of accuracy. However, they did not demonstrate how much they can increase the pre-warning time, which is an important indicator to assess performance of FCW systems. There is a work [7] that compared the system warning signals and the signals of driver's brake in order to show that they can increase the pre-warning time. They developed a FCW system based on THASV-II platform. The result demonstrated that the system warnings correspond to driver's brake behavior without measurement of the pre-warning time.

Our proposed idea to enhance FCW system is adding more functionalities of the warning to increase the pre-warning time. This can be done by analyzing the characteristic of collision and divide it into five levels. We implement a system that operates on multilevel of safety, which will help increase the performance. As mentioned above, the literature work attempted to predict frontal collisions and demonstrated their system's performance in the form of accuracy. In our work, we use warning lag time as a performance metric (section IV) because FCW requires a responsive vehicle maneuver in a short period of time to avoid accidents. The increase of pre-warning time in just a fraction of a second can mean life or death. Therefore, pre-warning time is a crucial property, in which we chose to be our performance indicator.

This paper is organized as follows. The concept of general FCW systems is described in section II. The proposed improvement of FCW system is presented in

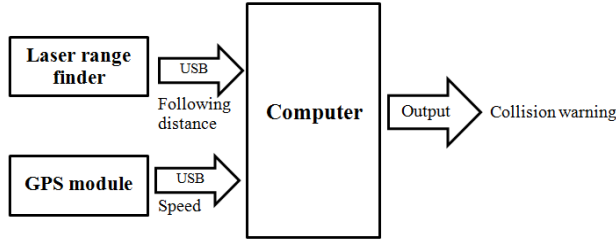


Figure 1. Forward collision warning system (FCW).

a method to improve FCW system. The warning lag time evaluation, a key performance of this work, is demonstrated in section IV. The systems were tested in real driving situation and the results are presented in section V. The conclusion is explained in section VI.

II. FORWARD COLLISION WARNING (FCW) SYSTEM

A forward collision warning system is an active pre-crash vehicle safety system. FCW informs the driver of possible frontal collisions to help the driver be aware of accidents and perform collision avoidance maneuvers. Generally, most of the FCW systems have been developed based on a kinematic model of collision in order to warn a driver of a potential crash when the distance to the front car is closer than the distance required for safe brake. The model involves equations of motion, which included distances, velocities, and accelerations. Many types of sensors are utilized to obtain distance to the front car, relative velocity to the car in front, the car's velocity, acceleration, etc. in order to construct a vehicle kinematic model and estimate frontal collision risks.

In our work, the FCW system we intended to improve is based on a basic formula of a vehicle kinematic model as shown in (1). We calculate the estimated time to the car in front (T_s) from a current position of a host car to a front car by using a current velocity of the host vehicle (v) and a current following distance (D). The vehicle's acceleration and the velocity of obstacle/car in the front are assumed to be zero in order to make the model as simple as possible.

The warnings are determined based on the two-second rule, which is a concept from defensive driving technique. The rule suggests that a driver has to keep a following distance at least 2 seconds from the car in front. The warning is active when T_s is less than this 2-second threshold. However, we found that 2 seconds were not suitable for roads in Bangkok, which has quite aggressive driving habit. Therefore, we reduced the threshold down to 1.6 seconds. This number came from trial and error in real driving experiments, in which we found that this value is the most suitable. The system computes the T_s value repeatedly according to the formula below, and gives a warning if the value is less than the threshold.

$$T_s = \frac{D}{v} \quad (1)$$

III. IMPROVEMENT OF FORWARD COLLISION WARNING

Since the average pre-warning time of typical FCW systems is quite short, there are efforts to increase this pre-warning time. We used another performance metric called

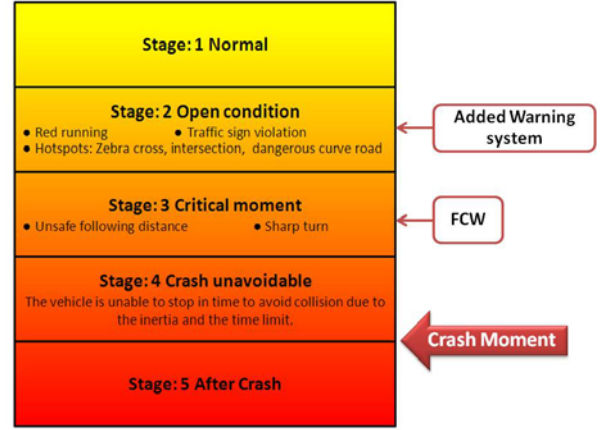


Figure 2. Five levels for crash prevention with the safety level of FCW system and the added warning system.

warning lag time which is directly related to the pre-warning time. We treat the warning lag time as the key performance of our FCW system. We propose that the improvement of FCW by increasing warning lag time can be performed in two ways. The first method, proposed in this paper, can be implemented by adding multilevel safe drive capability. The levels are divided into 5 levels according to collision risk. The second method is our future work, which will improve FCW systems by using a communication channel to promote cooperative driving among several vehicles. This idea is the implementation of distributed sensing in order to increase the range of sensors.

A. Multilevel of crash prevention

In our work, car crash event is divided into five levels according to collision risk as illustrated in Fig. 2. The minimal collision risk is classified as the first level. In this level, the driver pays attention while driving and follows traffic laws strictly. The possibility of accidents is very low. From the first level, if the driver places himself in an open condition to collision, he will be in the second level. The open condition to collision is usually caused by the violation of traffic regulation. In the second level, if the driver does not take a suitable maneuver to recover the driving situation back to the first level, the driving situation will turn into the third level which is a critical moment of crash avoidance. In the third level, the driver can avoid collision if he has good skills of maneuver such as harsh brake or crisp turn. However, when the crash becomes unavoidable, the moment before crash is classified as the fourth level. The fifth level is the time period after crash.

From five levels of crash prevention, typical FCW systems operate on stage 3, the critical moment. It is the moment about 2-3 seconds before the impact. In order to increase pre-warning time, we implemented FCW system according to the multilevel of safety based on assumption that the pre-warning time can be increased by combining FCW system with the warning system that operates on stage 2, open condition, in order to prevent a driver getting into stage 3, critical moment. We chose the location-based warning system (LW) for this purpose. The location-based warning operates at stage 2 because it will warn the driver when the driver drives toward a dangerous spot such as a

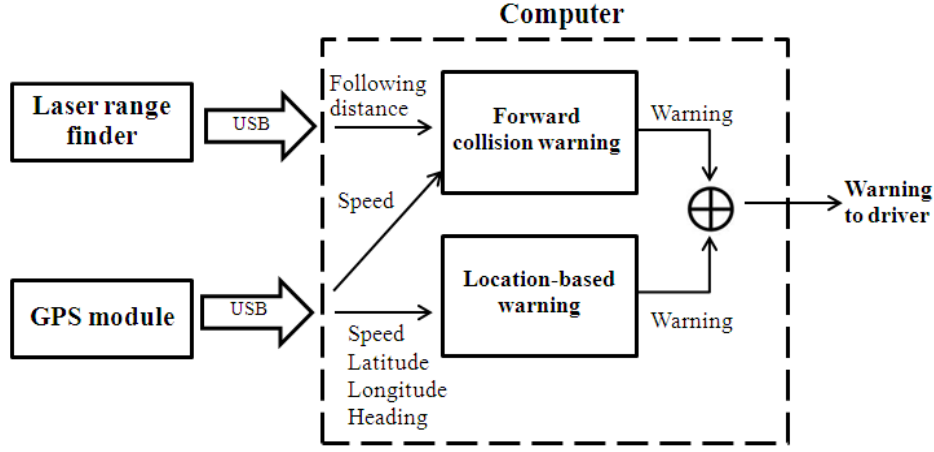


Figure 3. Combination of FCW and LW systems.

intersection or a sharp curve with high speed. The vehicle may go pass the intersection without a problem if there is no car around. However, this move has a high risk. If there is another car approaching from the side, the situation will turn from stage 2 into stage 3 and the driver may have to apply sharp brake later on. Therefore, the warning when approaching an intersection is considered as warning on stage 2 because the driver has placed himself into an open condition by driving too fast.

B. Combination of FCW and LW system

We chose location-based warning to enhance FCW because it is the most reliable system that works on stage 2. The main reason is that the warnings generated are from static environment. Location-based warning system informs a driver about potentially dangerous places (called accident hotspots) before reaching them in order to help a driver become aware of the dangers and prepare to slow down before reaching the hotspots. The accident hotspots are dangerous places which possibly lead to accidents such as a zebra cross, an intersection, a dangerous curve road, etc.

We combined FCW and LW systems by summing up the warning outputs from both systems. The two systems run separately and concurrently as illustrated in Fig. 3. Two sensors were utilized and connected to a computer to obtain driving information for FCW and LW. A laser rangefinder was used to measure a following distance. A GPS module was used to obtain a vehicle status such as speed, vehicle position (latitude and longitude) and heading. For FCW, the following distance and the speed were used to determine a warning as described in section II. For LW, a warning was calculated from vehicle's speed, vehicle's position and vehicle's heading according to the following equation.

$$LW = \begin{cases} 1 & , (d < d_{th}) \text{ AND } (v > v_{th}) \text{ AND } (|h - h_D| < h_{th}) \\ 0 & , \text{otherwise} \end{cases} \quad (2)$$

For the LW system that we used, all accident hotspots were collected in a database. Each data point consists of latitude, longitude, speed limit and heading. The car status from GPS will be compared to every data point nearby. If the location is close enough, the current velocity (v) exceeds the speed limit (v_{th}), and the heading (h) points toward a

dangerous place, then the system warning will be issued. For the current location-based warning system shown in (2), the speed limit (v_{th}), and the range of warning distance from hotspots (d_{th}) are 30km/h and 30 m. respectively. h_D is the angle from vehicle position, used as an origin point, to the nearest hotspot in the database. If an absolute difference of h and h_D is not over h_{th} , it means that the car is moving toward the hotspot. As shown in Fig. 4., we differentiate a vehicle direction that it is going into or going out of a hotspot by considering a vehicle's heading (h) and an angle from vehicle position to the nearest hotspot in the database (h_D). When a vehicle is going into a hotspot, h and h_D are same direction. In contrast, h and h_D are opposite direction when a vehicle is going out of a hotspot.

IV. WARNING LAG TIME FOR PERFORMANCE EVALUATION

Since FCW systems are activated at a critical moment, within 2-5 seconds before accidents, timing is a crucial parameter. Some researchers focused their study on the pre-warning time, which is defined as the time period between the issue of warning and the moment of crash impact. They assumed that pre-warning time can be measured, because crashes can be emulated in a driving simulator. Our work, which focuses on real driving situations and experiments, cannot simulate crashes. Therefore, we proposed another performance metric, the warning lag time as shown in Fig. 5 and 6, which is used as a performance indicator in our experiments. The warning lag time is defined as the time period between the issue of warning and the execution of brake by a normal driver who has no active warning systems. In typical FCW systems, the warnings usually come after brakes from alert drivers because human perception is better than current sensing technology. As a result, human drivers can perceive more road environments and react quicker than a machine.

We evaluated the warning lag time of our system based on the assumption that the pattern of system warning signal is similar to the pattern of actual brake signal. However, the system warning has a short time-shift from the actual brake. Thus, this short time-shift is warning lag time of the system warning. After obtaining both data, we can evaluate warning lag time by following these steps. First, we calculate the cross-correlation between the actual brake and the system

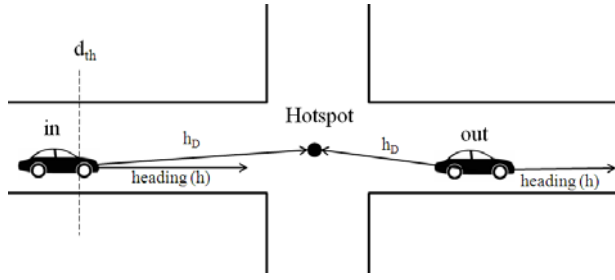


Figure 4. Warning condition of Location-based warning system.

output. The second step is finding the local maxima around the middle of cross-correlation output. This is because the time lag will not deviate much from the center (delay = 0 sec). Finally, we get the warning lag time by measuring a distance between the center of cross-correlation output and the local maxima as shown in Fig. 7. This figure demonstrates the measurement of warning lag time by a cross-correlation result of the actual brake and the system warning of FCW system with LW system. The vertical line indicates the center of cross-correlation output (no shift). The local maxima point is on the right of the center. This means that the system warning happens before the actual brake with the measured pre-warning time. On the other hand, if the local maxima point is on the left, it indicates that the system warning delays from the actual brake.

V. EXPERIMENTS AND RESULTS

Our system consists of three sensors installed in a test vehicle as shown in Fig. 8. The first sensor is a laser rangefinder mounted at the car's windshield and pointed forward. This sensor is used to obtain the distance between the test car and a car in the front. The second sensor is a GPS receiver used to obtain the car's status consisting of latitude, longitude, speed and heading of the car on the earth. The last one is a force sensing resistor (FSR) equipped on a brake pedal to obtain actual brake actions of the driver. The force sensor is used only for performance evaluation purpose. It is not needed for normal operation of the warning system. These sensors are connected to a main computer in order to record and process the data.

The purpose of the experiments is to demonstrate that the warning lag time can be increased by adding the functionality of a FCW system with a location-based warning system. We tested both systems, the FCW system with and without Location-based warning system, in real driving situation for 30 minutes on city roads. The test route consists of 28 accident hotspots. For testing purpose, the driver does not know the output of the systems, so that his driving behavior is independent to the output from the systems.

For performance evaluation, we determined two performance parameters which are the warning lag time and the accuracy of system warning. The accuracy of system warning is the comparison result between the system warnings and the actual brakes as shown in Fig. 9. After determining the warning lag time, the time-scale of system warnings was shifted by the warning lag time before calculating the accuracy in order to measure the system warnings corresponding to the actual brakes.

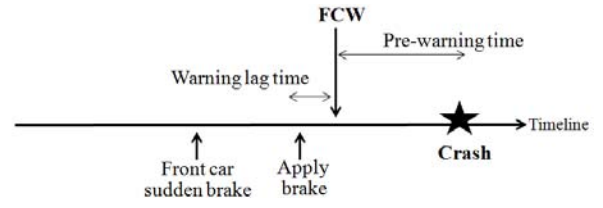


Figure 5. The pre-warning time and the warning lag time in a timeline of driving situation.

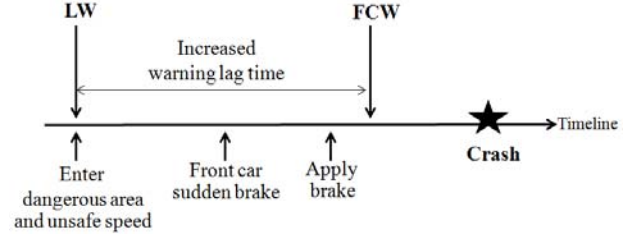


Figure 6. The increased warning lag time when enhancing FCW by adding LW.

From Table I., the accuracy of the combination FCW+ LW system decreases from the accuracy of FCW alone for about 0.53%. The main cause of false negatives of FCW in this test is the traffic congestion of the road that we performed the test. When the car stopped, the driver still left his foot on the brake pedal. This was counted as a brake action, while the FCW system did not give any warning when the car did not move. The output from FCW system has an average delay of 1.47 sec. from the actual brake, whereas the warning lag time of the combination system of FCW and LW is 1.32 sec. These results indicate that enhancing FCW system with location-based warning system can help improve the pre-warning time for 0.15 sec. (from the pre-waning time +1.47 sec. to the pre-warning time +1.32 sec.).

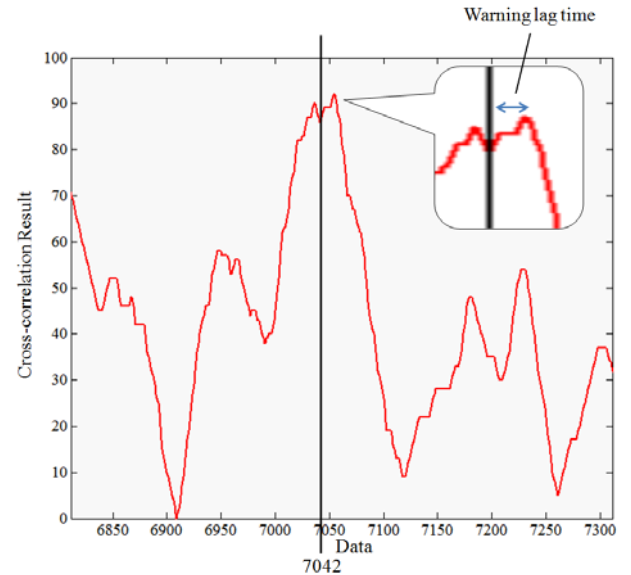


Figure 7. Measurement of warning lag time from cross-correlation between actual brakes and system output.

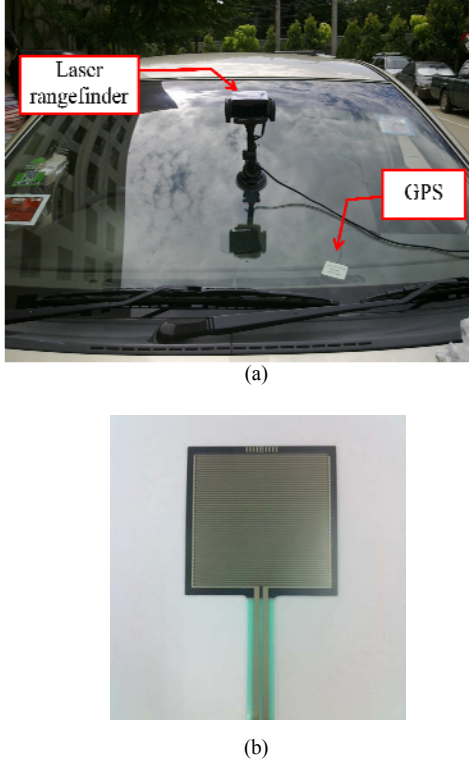


Figure 8. System Components: (a) Laser rangefinder and GPS receiver, and (b) Force Sensing Resistor for obtaining actual brakes of the driver

TABLE I. THE RESULTS OF PERFORMANCE EVALUATION OF THE FCW SYSTEM WITHOUT AND WITH LOCATION-BASED WARNING SYSTEM

Warning System	Warning lag time (s)	Accuracy of system (%)		
		Correct	False positive	False negative
FCW	+1.47	79.54	10.09	10.37
FCW with LW	+1.32	79.01	10.66	10.33

(-) and (+) means the warning lag time happens before and after the actual brake respectively

VI. CONCLUSION

In this work, we improved the performance of FCW system to increase the warning lag time by adding the location-based warning system which is in the second level of the crash prevention hierarchy. In order to perform our system in real experiments, we proposed the warning lag time as the key performance which is a time period between an actual brake and a warning signal. We calculated the warning lag time based on correlation method as shown in section 3. The systems, FCW without and with LW, were tested in real driving situations in order to evaluate the performance of the enhanced FCW system. The results show that enhancing normal FCW system with the level 2 warning system can help increase the pre-warning time by 0.15 sec. with a small drop in accuracy.

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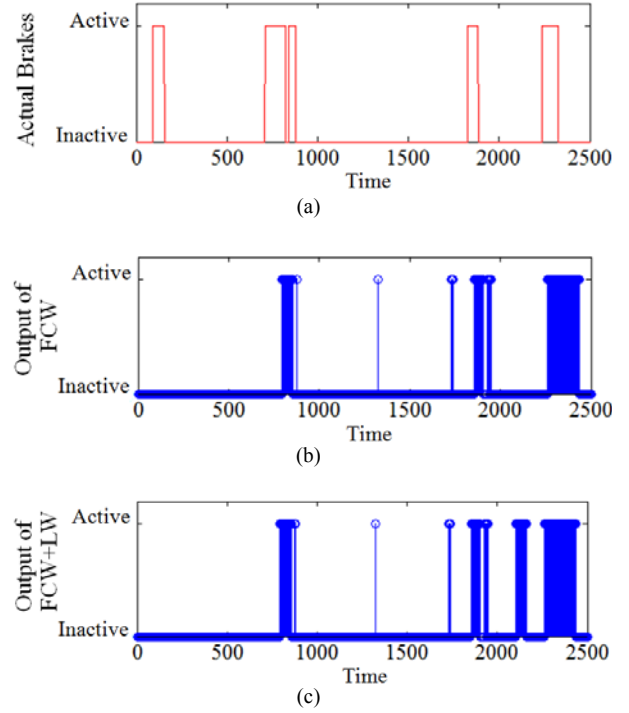


Figure 9. The comparison of (a) the actual brakes with (b) the output of the FCW system and (c) the output of the combination of FCW and LW systems..

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